ScienceX

Grid-Scale Battery Storage Monthly Report, October 2026

Science X Inc. Published November 1, 2026

The price of electricity changes with the time of day. "Grid-scale battery storage" (large batteries connected to the transmission grid that store electricity and release it) earns money from that difference. This report uses published figures and our own records to show this month's trends in figures.

1. Three things that happened this month

Wholesale electricity is the electricity that power companies and retailers buy and sell. Its price is set on the "spot market" of the Japan Electric Power Exchange (JEPX), where electricity for the next day is bought and sold the day before.

- In October 2026, the average wholesale electricity price was 27.02 yen/kWh in Tokyo (13.06 yen/kWh in the same month last year) and 15.07 yen/kWh in Kyushu (10.74 yen/kWh in the same month last year).
- The monthly average spread between the highest 4 hours and the lowest 4 hours was 17.14 yen/kWh in Tokyo (7.11 yen/kWh in the same month last year) and 19.30 yen/kWh in Kyushu (9.23 yen/kWh in the same month last year).
- In September 2026 (preliminary), the average awarded price for Primary reserve was 3.43 yen/kW per 30 min, and batteries made up 27.9% of it.



Figure 1 Average wholesale electricity price (yen/kWh). Navy is October 2026, grey is the same month last year

Source: Spot market trading results published by the Japan Electric Power Exchange (JEPX). Simple average of the price for each 30 minutes. Our calculation.

2. How the electricity price moves during the day

The electricity price changes every 30 minutes. The figure shows the shape of a day, averaging the price at the same time across the month. If a battery stores electricity when it is cheap and releases it when it is expensive, the difference becomes its income.

The cheapest time in both Tokyo and Kyushu was around 12:00 (19.03 yen in Tokyo, 4.94 yen in Kyushu). The most expensive was 37.11 yen around 16:30 in Tokyo and 24.74 yen around 17:30 in Kyushu.

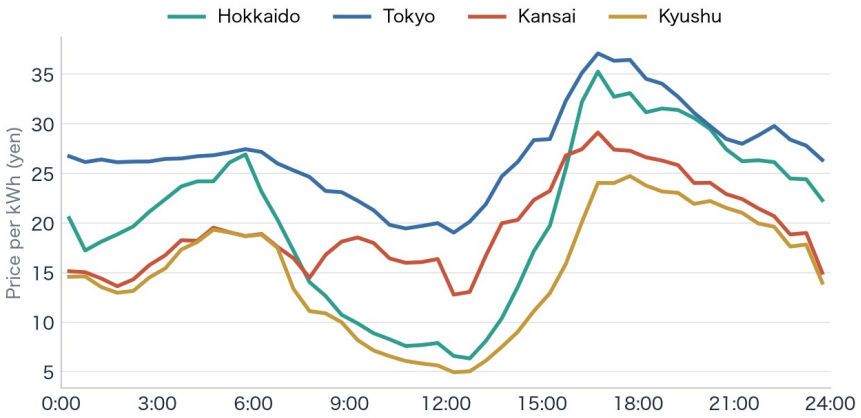


Figure 2 Daily price peaks and troughs (October 2026 average, yen/kWh)

Source: JEPX spot market. Our calculation.

3. How much is the price spread per kWh

A battery buys and stores electricity when it is cheap and sells it when it is expensive. The figure shows, for each area, the monthly average of the day's highest 4-hour average price minus the day's lowest 4-hour average price. This equals the difference per kWh if you buy 1 kWh in the cheapest 4 hours and sell it in the most expensive 4 hours. It is a simple difference: it does not subtract electricity lost in charging and discharging or fees, and it ignores the order of the cheap and expensive hours.

The largest monthly average spread between the highest 4 hours and the lowest 4 hours was 29.02 yen/kWh in Hokkaido, and the smallest was 17.14 yen/kWh in Tokyo.

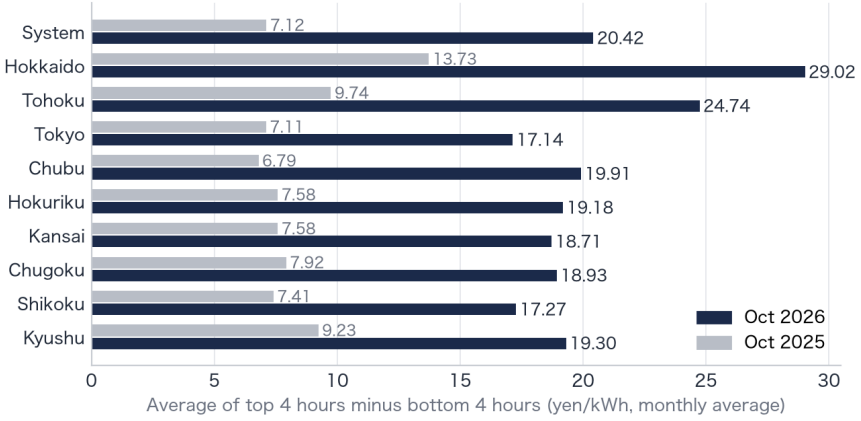


Figure 3 Average of the highest 4 hours − average of the lowest 4 hours (yen/kWh, monthly average). Navy is October 2026, grey is the same month last year

Source: JEPX spot market. Our calculation.

Area	2-hour spread (yen/kWh)	4-hour spread (yen/kWh)	4-hour, same month last year (yen/kWh)	Days with a 4-hour spread of 10 yen or more
System	22.95	20.42	7.12	29
Hokkaido	32.56	29.02	13.73	31
Tohoku	27.09	24.74	9.74	30
Tokyo	19.62	17.14	7.11	29
Chubu	22.37	19.91	6.79	29
Hokuriku	21.71	19.18	7.58	31
Kansai	21.22	18.71	7.58	31

© Science X Inc. · https://www.scix.co.jp

Area	2-hour spread (yen/kWh)	4-hour spread (yen/kWh)	4-hour, same month last year (yen/kWh)	Days with a 4-hour spread of 10 yen or more
Chugoku	20.90	18.93	7.92	31
Shikoku	19.40	17.27	7.41	28
Kyushu	21.19	19.30	9.23	31

The spread is the monthly average of the daily spreads. October 2026 covers 31 days.

4. There is a spread every day

We lined up the 4-hour spread day by day. The larger the spread on a day, the more room a battery has to earn from it.

Out of 31 days, the 4-hour spread was 10 yen or more on 29 days in the System, 29 days in Tokyo and 31 days in Kyushu. The area with the fewest such days was Shikoku, with 28 days.

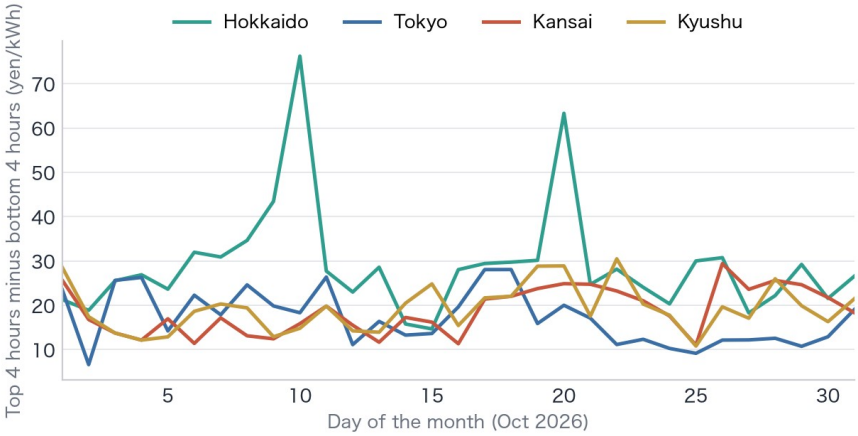


Figure 4 Daily spread (the day's highest 4-hour average − lowest 4-hour average, yen/kWh)
Source: JEPX spot market. Our calculation.

5. Three income sources for a battery storage site

A battery storage site has three main sources of income: the price spread (wholesale electricity market), the balancing market (a market that buys and sells the "capacity" to fix a surplus or shortage of electricity) and the capacity market (a market that buys future electricity supply capacity in advance). In the capacity market, power plants and storage sites earn a fixed income per kW.

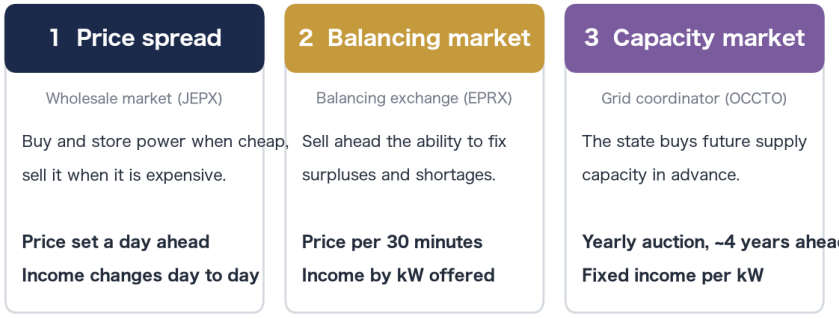


Figure 5 Three income sources for a battery storage site
Source: Rules of the Japan Electric Power Exchange (JEPX), Electric Power Reserve Exchange (EPRX) and Organization for Cross-regional Coordination of Transmission Operators (OCCTO).

6. The six products of the balancing market

The balancing market is a market where grid operators buy, every 30 minutes, the "capacity" to fix a surplus or shortage of electricity. Products are divided by how fast output must change after an instruction. Primary reserve is the fastest and Tertiary reserve 2 is the slowest. A Composite product is a bid that offers several products together. Prices are set in yen per 1 kW of capacity offered per 30 minutes. Published figures for October are not yet in our data, so we show September 2026 (preliminary), the latest month with all products.

In September 2026 (preliminary), the volume bid was larger than the volume sought for every product, with bid ratios from 123.1% to 377.3%. The lowest award rate was 26.9% for Tertiary reserve 2.

Product	Time from instruction to output change	Duration
Primary reserve	Within 10 seconds	5 minutes or more
Secondary reserve 1	Within 5 minutes	30 minutes
Secondary reserve 2	Within 5 minutes	30 minutes
Tertiary reserve 1	Within 15 minutes	30 minutes
Tertiary reserve 2	Within 60 minutes	30 minutes

Figures for September 2026 (preliminary, nationwide).

Product	Volume sought (MW)	Volume bid (MW)	Volume awarded (MW)	Bid ratio (bid ÷ sought)	Award rate (awarded ÷ bid)	Average awarded price (yen/kW per 30 min)	Battery share
Primary reserve	1,388.7	1,710.0	1,337.3	123.1%	78.2%	3.43	27.9%
Secondary reserve 1	962.6	1,996.2	1,102.3	207.4%	55.2%	2.49	1.8%
Secondary reserve 2	983.1	3,709.3	2,163.2	377.3%	58.3%	2.51	1.0%
Tertiary reserve 1	2,498.5	5,895.9	3,021.0	236.0%	51.2%	2.28	0.7%
Tertiary reserve 2	376.3	1,371.7	368.9	364.5%	26.9%	2.22	0.8%
Composite product	3,383.8	6,641.4	3,623.5	196.3%	54.6%	2.64	10.4%

Volumes are averages per 30-minute slot. A product with a bid ratio below 100% has fewer bids than the volume sought. The award rate is the share of the volume bid that was awarded.

Source: Trading results (preliminary) and trading results by power source type published by the Electric Power Reserve Exchange (EPRX). Our calculation.

© Science X Inc. · https://www.scix.co.jp

7. How the price of each product has moved

This shows the average awarded price for each product since April 2025 (monthly average weighted by volume awarded). From March 14, 2026, Primary reserve through Tertiary reserve 1 and the Composite product moved from trading every 3 hours to trading every 30 minutes (Tertiary reserve 2 was already every 30 minutes).

The average awarded price for Primary reserve went from 2.89 yen in April 2025 to 3.43 yen in September 2026, and the highest was 4.90 yen in February 2026 (yen/kW per 30 min).

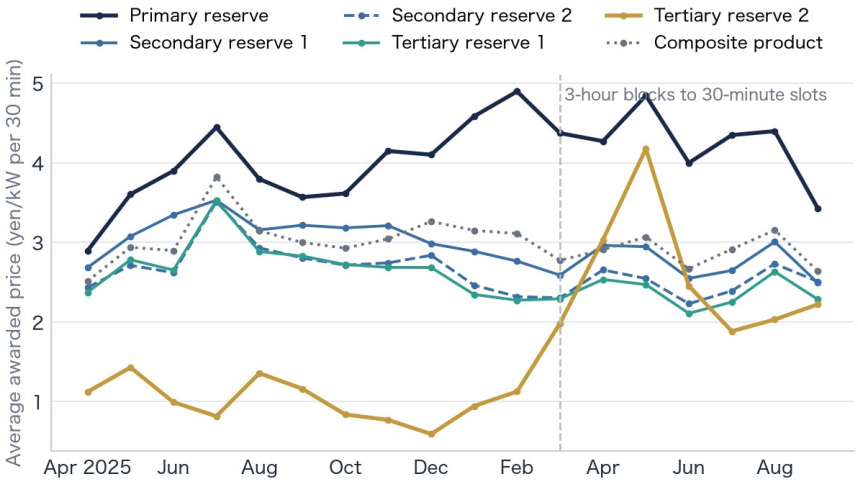


Figure 6 Average awarded price by product in the balancing market (yen/kW per 30 min, monthly)
Source: Trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our calculation.

8. Which hours of the day are expensive

This shows the average awarded price for each product, hour by hour, in September 2026.

In September 2026, the average awarded price for Primary reserve was highest around 18:00 at 4.09 yen and lowest around 0:00 at 2.75 yen (yen/kW per 30 min).

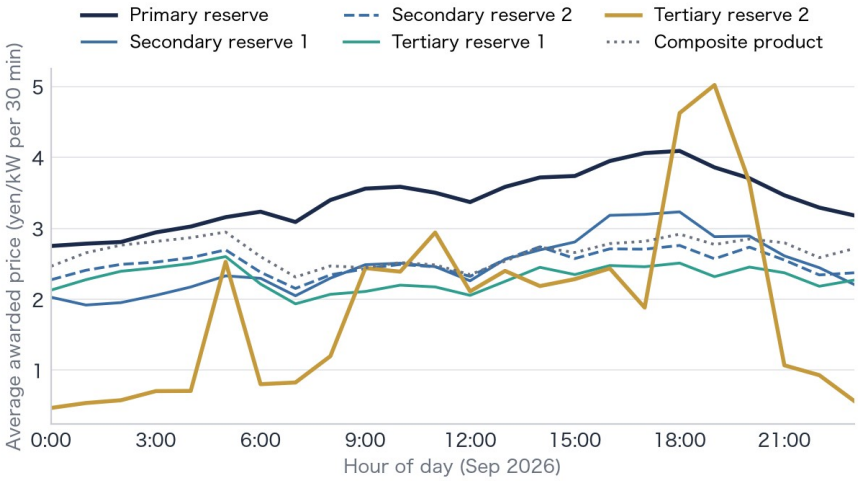


Figure 7 Average awarded price by hour (September 2026, yen/kW per 30 min)
Source: Trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our calculation.

9. Which products batteries are used for

The figure shows the share of the awarded volume taken by batteries. The table compares the average awarded price of the volume awarded to batteries with the average awarded price of the whole product.

In September 2026, batteries made up 27.9% of Primary reserve, and the average awarded price of the volume awarded to batteries was 5.17 yen (3.43 yen overall, yen/kW per 30 min).

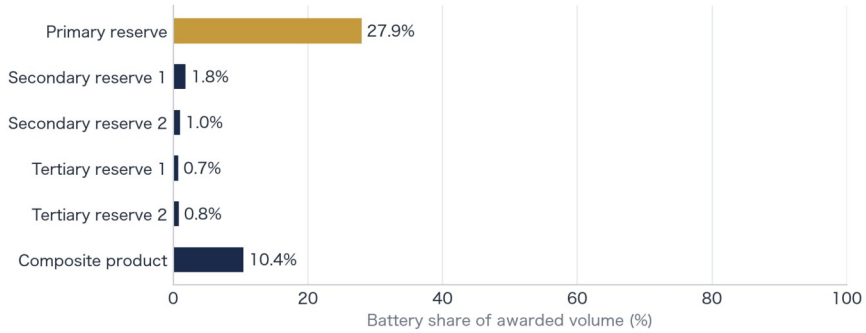


Figure 8 Share of batteries in the balancing market (%)

Source: Trading results (preliminary) by power source type published by the Electric Power Reserve Exchange (EPRX). Our calculation.

Product	Volume awarded to batteries (yen/kW per 30 min)	Whole product (yen/kW per 30 min)
Primary reserve	5.17	3.43
Secondary reserve 1	2.18	2.49
Secondary reserve 2	1.85	2.51
Tertiary reserve 1	1.86	2.28
Tertiary reserve 2	59.26	2.22
Composite product	5.13	2.64

Both are averages weighted by volume awarded (September 2026, preliminary, nationwide).

10. Tertiary reserve 2 is concentrated in high-price slots

Tertiary reserve 2 is a product that covers forecast errors in renewable power output caused by the weather. Its price is low in most 30-minute slots and high in only a very few.

In September 2026, out of 1,194 slots with awards, the top 1% of slots by revenue (average awarded price × volume awarded) made up 9.4% of the month's revenue. There were 25 slots where the nationwide average awarded price was 10 yen/kW per 30 min or more, and 198 slots with no awards.

Slots with the highest maximum awarded price (by area where the power source is located):

Date	Time	Area	Maximum awarded price (yen/kW per 30 min)
September 8	9:30	Tokyo	96.50
September 8	10:30	Tokyo	96.50
September 8	11:00	Tokyo	96.50
September 27	11:00	Tokyo	96.00
September 27	11:30	Tokyo	96.00

Source: Tertiary reserve 2 trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our calculation.

11. The capacity market buys future supply capacity

The capacity market buys electricity supply capacity for 4 years ahead in a once-a-year auction, and the price is set in yen per kW per year. The left figure shows the clearing prices of past main auctions. The right figure shows the capacity of batteries and pumped hydro contracted in the Long-Term Decarbonization Power Source Auction (a separate auction that supports new batteries, pumped hydro and similar). These are published once a year, so they do not change every month. The latest is delivery fiscal year FY2029 (published January 2026).

In the latest once-a-year figures, the capacity market price for Tokyo was 15,111 yen/kW for FY2029, and in the Long-Term Decarbonization Power Source Auction FY2025 bids, batteries and pumped hydro were 819 MW of lithium-ion and 886 MW of non-lithium.

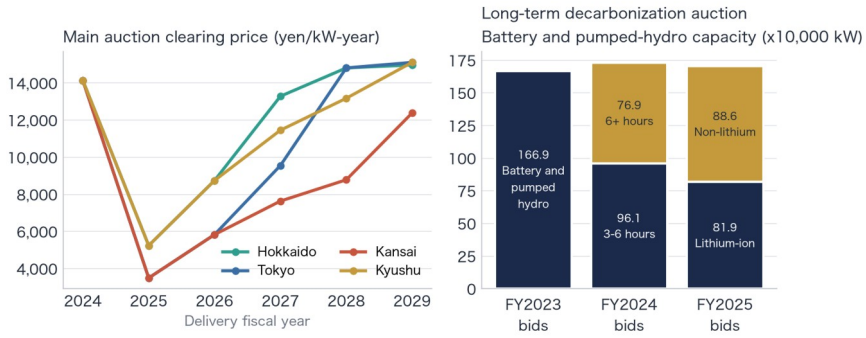


Figure 9 Capacity market clearing price (left, yen/kW) and contracted capacity of batteries and pumped hydro in the Long-Term Decarbonization Power Source Auction (right, in units of 10,000 kW)

Source: Auction results and capacity securing contract documents published by the Organization for Cross-regional Coordination of Transmission Operators (OCCTO) (links in section 15). As published.

12. Where battery storage projects are growing

We counted, by grid area of the grid operator, the projects we newly listed this month on our DealRoom (our data room for buyers), a site where we list battery storage projects for buyers. These are figures for projects we handled, not the number for the whole market. Transfer of rights is a sale that hands over the rights to the land and the grid connection. Completed-build handover is a sale that hands over the site once the equipment is built.

In October 2026, there were 77 new projects totaling 309.1 MW. By grid area, Tokyo had 33 projects and 99.8 MW, and Tohoku had 8 projects and 73.4 MW.

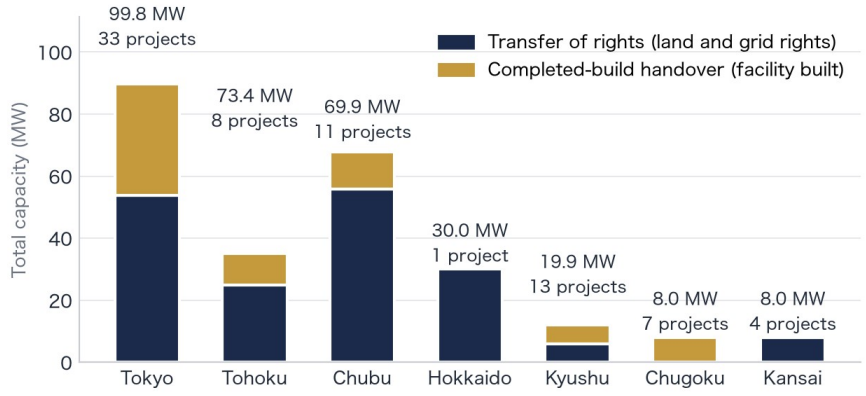


Figure 10 Output (MW) of projects we listed on our DealRoom in October 2026

Source: Our DealRoom records (project locations are given only to the grid area, and prices and parties are not included).

13. How many months and how much it takes to connect to the grid

A grid connection study is the procedure in which the grid operator checks "whether this location can connect, and how much the work will cost." The grid connection cost contribution is the part of the connection work cost that the applicant pays. Of the responses we received, we counted only high-voltage ones (connections to a 6,600V distribution line), by grid area. Extra-high voltage is not included. Grid areas with fewer than 5 responses are not shown.

For high voltage only (as of October 19, 2026), the median construction period was 8.0 months in Tokyo and 17.0 months in Tohoku, and the median grid connection cost contribution was ¥3.02 million in Tokyo and ¥32.35 million in Tohoku.

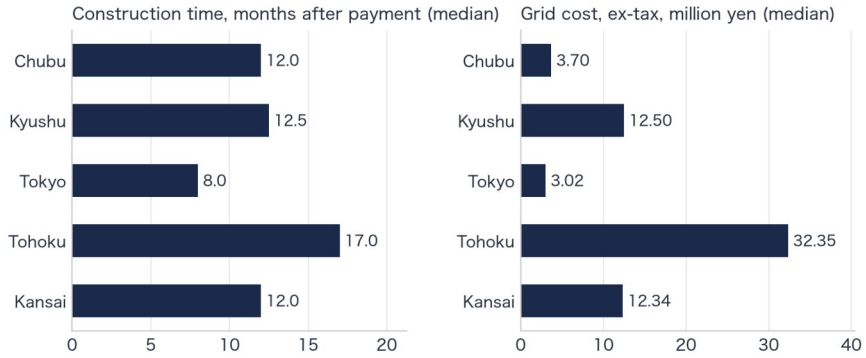


Figure 11 Median grid connection study responses (high voltage) by grid area (left: months of construction work, right: grid connection cost contribution, excluding tax)

Source: Our calculation from high-voltage grid connection study responses from general transmission and distribution operators that we received (as of October 19, 2026).

14. Market figures (tables you can use as they are)

Of the figures behind the charts in this report, we list the wholesale electricity (JEPX) figures by area, trading volumes, and capacity market clearing prices. The figures for each balancing market product are in the table in section 6.

JEPX spot market (October 2026, by area, yen/kWh)

Area	Monthly average	Same month last year	Daytime (8:00–16:00)	Evening (16:00–20:00)	4-hour spread	Highest	Slots at 30 yen or more	Slots at 0.01 yen
System	22.51	11.67	16.02	31.06	20.42	47.88	220	39
Hokkaido	20.49	12.89	11.31	32.25	29.02	105.00	303	67
Tohoku	23.99	12.08	15.78	34.19	24.74	50.01	437	95
Tokyo	27.02	13.06	23.12	34.69	17.14	50.01	466	16
Chubu	25.46	11.68	20.38	33.55	19.91	46.45	387	24
Hokuriku	19.70	11.27	18.29	26.92	19.18	46.45	218	24
Kansai	19.50	11.27	18.22	26.76	18.71	46.45	215	29
Chugoku	15.63	11.12	9.86	23.10	18.93	44.79	63	79
Shikoku	11.44	9.28	7.50	16.61	17.27	37.51	23	146
Kyushu	15.07	10.74	8.31	23.10	19.30	44.79	53	91

Trading volume (October 2026, total for 31 days): total contracted volume 30,319 million kWh (22,279 million kWh in the same month last year), sell bids 45,218 million kWh, buy bids 38,191 million kWh.

Source: Spot market trading results published by the Japan Electric Power Exchange (JEPX). Prices are simple averages of 30-minute slots. Our calculation.

Capacity market main auction area prices (yen/kW per year)

Delivery fiscal year	Hokkaido	Tohoku	Tokyo	Chubu	Hokuriku	Kansai	Chugoku	Shikoku	Kyushu
FY2024	14,137	14,137	14,137	14,137	14,137	14,137	14,137	14,137	14,137
FY2025	5,242	3,495	3,495	3,495	3,495	3,495	3,495	3,495	5,242
FY2026	8,749	5,833	5,834	5,832	5,832	5,832	5,832	5,832	8,748
FY2027	13,287	9,044	9,555	7,823	7,638	7,638	7,638	7,638	11,457
FY2028	14,812	14,812	14,812	10,280	8,785	8,785	8,785	8,785	13,177
FY2029	14,972	15,111	15,111	12,388	12,388	12,388	12,388	12,388	15,112

Contracted capacity of batteries and pumped hydro in the Long-Term Decarbonization Power Source Auction (10,000 kW)

Bid fiscal year	Category	Contracted capacity (10,000 kW)
FY2023 bids	Batteries and pumped hydro	166.9
FY2024 bids	3 to 6 hours	96.1
FY2024 bids	6 hours or more	76.9
FY2025 bids	Lithium-ion	81.9
FY2025 bids	Non-lithium	88.6

Source: Auction results and capacity securing contract documents published by the Organization for Cross-regional Coordination of Transmission Operators (OCCTO) (links in section 15). As published.

15. Assumptions and terms

- The figures are our calculations from the published values of the sources. We have not added up several months or converted to annual figures. We do not show revenue amounts for any single storage site.
- Months covered: wholesale electricity (JEPX) is October 2026 (31 days), and the balancing market (EPRX) is September 2026 preliminary values. The balancing market will cover a different month until published figures for October are in our data.
- Average prices are simple averages of 30-minute slots (not weighted by trading volume). Daytime is the slots from 8:00 to 16:00, and evening is the slots from 16:00 to 20:00.
- Spread: the day's 30-minute slots are sorted by price, and the average of the 8 cheapest slots is subtracted from the average of the 8 most expensive slots (4 hours) (4 slots for 2 hours). It does not consider the order of the cheap and expensive hours, electricity lost in charging and discharging, or fees, taxes and grid usage charges. It does not show actual revenue.
- Balancing market: nationwide figures (the total column) from EPRX preliminary values. The volume sought is by grid operator (TSO), and the volume bid, volume awarded and average awarded price are by the area where the power source is located. Volumes are averages per 30-minute slot. The average awarded price is weighted by volume awarded. Bid ratio = volume bid ÷ volume sought; award rate = volume awarded ÷ volume bid. For the 3-hour blocks until March 13, 2026, we counted 1 block as 6 30-minute slots.
- The time from instruction to output change and the duration for each balancing market product follow the FY2026 specifications.
- High-price slots for Tertiary reserve 2: slots where the nationwide average awarded price was 10 yen/kW per 30 min or more. The top 1% are the top 1% of slots sorted by revenue (average awarded price × volume awarded). The maximum awarded price is the published value by area where the power source is located.

Sources (OCCTO, capacity market auction results): delivery fiscal year FY2024 https://www.occto.or.jp/iinkai/youryou/kentoukai/2020/files/youryou_kentoukai_27_05.pdf / FY2025 https://www.occto.or.jp/iinkai/youryou/kentoukai/2021/files/youryou_kentoukai_35_03_1.pdf / FY2026 https://www.occto.or.jp/iinkai/youryou/kentoukai/2022/files/youryou_kentoukai_43_03.pdf / FY2027 https://www.occto.or.jp/market-board/market/oshirase/2023/files/240124_mainauction_youryouyakujokekka_kouhyou_jitsujukyu2027.pdf / FY2028 https://www.occto.or.jp/market-board/market/oshirase/2024/files/250129_mainauction_youryouyakujokekka_kouhyou_jitsujukyu2028.pdf / FY2029 https://www.occto.or.jp/assets/news/capacity-market/260120_mainauction_youryouyakujokekka_kouhyou_jitsujukyu2029.pdf

Sources (OCCTO, Long-Term Decarbonization Power Source Auction): FY2023 bids https://www.occto.or.jp/assets/260605_auction_keiyakujyoukyou_kouhyou.pdf / FY2024 bids https://www.occto.or.jp/market-board/market/jitsujukyukanren/files/longauction_keiyakukekka_2024.pdf / FY2025 bids https://www.occto.or.jp/assets/various/capacity-market/jitsujukyukanren/yoryokakuhokeiyaku/longauction_keiyakukekka_2025.pdf

- Capacity market auction results for each delivery fiscal year are published in a different year (FY2024 in 2020, FY2029 in January 2026). The number of battery projects in the Long-Term Decarbonization Power Source Auction is not shown because it is not in the published summary tables.

Sources: Japan Electric Power Exchange JEPX spot market trading results / Electric Power Reserve Exchange EPRX trading results (preliminary) and trading results by power source type / Organization for Cross-regional Coordination of Transmission Operators (OCCTO) capacity market main auction results and Long-Term Decarbonization Power Source Auction capacity securing contract documents / our calculation from grid connection study responses we received / our DealRoom records.

Terms: the grid = the power companies' transmission network. Grid connection = connecting to the transmission network. MW = a measure of power (1MW = 1,000kW). MWh = the amount of electricity that can be stored. yen/kWh = the price per 1 kWh of electricity.

© Science X Inc. · https://www.scix.co.jp

Science X Inc. https://www.scix.co.jp Contact s@scix.co.jp

Newsletter "Japan grid-scale BESS: primary data":
https://www.linkedin.com/newsletters/7521830530569445376/

You may use the figures and numbers if you cite the source (citation guide https://www.scix.co.jp/press).